Benton Tripp

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Experienced data scientist with a background in data analytics, machine learning, statistical modeling, and geospatial analytics. I bring over six years of professional experience across various industries, specializing in the development and implementation of analytics-driven solutions. Known for my self-motivation, strong communication skills, and ability to learn and apply new methodologies efficiently, I strive to deliver innovative insights and exceed expectations in both team-oriented and independent settings.

Work Experience

Highly skilled in technical problem-solving and data analysis, I leverage my expertise in machine learning, statistical modeling, and geospatial technologies to support business objectives. My technical competencies span advanced R and Python programming, cloud computing (AWS, Azure), and deployment of machine learning pipelines using MLOps platforms. As a natural leader and team player, I consistently deliver high-quality results, providing valuable insights and fostering a collaborative environment to drive success.

Technology Analyst

City of Raleigh
Raleigh, NC
June 2025 - Present

- Evaluate and assess department data systems and analytical processes to recommend improvements aligned with business goals, collaborating with cross-functional teams to deploy data-driven solutions.
- Built a computer vision application using PyTorch and a YOLO-based model to detect engineering seals and other sensitive information in large PDF plan sets and permit documents, enabling automatic redaction and significantly reducing manual review burden for the records team.
- Developed and maintain city population estimates and projections using statistical and spatial modeling methods, integrating census data, housing unit inventories, and permit activity to support planning decisions and long-range forecasting.
- Designed and developed public-facing Power BI dashboards for the Review Turnaround Times and Performance Dashboards webpage, enabling residents and developers to track permit and

site plan review times, benchmark compliance, and intake durations across residential and non-residential applications.
- Built interactive dashboards for the Blueprint for Development: An 8-Stage Framework, providing citywide performance metrics across the full development lifecycle—from comprehensive planning and annexations through building permits, inspections, and surety release.
- Created over 20 internal Power BI dashboards for operational data analyis and reporting.
- Conduct data analysis to identify areas for process enhancement, including the suburban and urban development trends analysis presented to City Council and senior leadership.
- Developed and maintain automated workflows to streamline department operations, including council meeting trackers, event calendars, and automated email processing for customer intake queues.
- Prepare reports and technical documentation; provide support and training to end-users on dashboard tools and data visualization best practices.

Data Scientist

Tennessee Valley Authority
Chattanooga, TN (Remote)
August 2022 - June 2025

- Developed end-to-end machine learning solutions, implementing CI/CD data pipelines, advanced statistical modeling, and interactive web applications using R Shiny for Groundwater Evaluation, Geochemical Modeling, and Billing Anomaly Detection systems.
- Led technical data science proof-of-concepts utilizing tools like Python, R, SQL, AWS, and Git to solve complex business problems.
- Built document analysis tools incorporating LLMs and Retrieval-Augmented Generation (RAG) systems using Python and LangChain, demonstrating expertise in modern NLP techniques.
- Participated in an enterprise MLOps platform assessment (Domino Data Lab, DataRobot, Dataiku, SAS, AWS SageMaker) for model monitoring and workflow optimization.
- Effectively communicated complex technical concepts to non-technical stakeholders through developing and instructing advanced R programming courses.
- Leveraged Azure DevOps, AWS services, and Dataiku to develop, deploy, and manage scalable machine learning pipelines, ensuring reliable model serving in production environments.

Data Analyst

Elder Research Inc.
Raleigh, NC
August 2021 - August 2022

- Engineered an ML pipeline using Databricks for ETL, model training, and deployment that identified major slippage drivers for Church & Dwight, resulting in over $1 million in savings through predictive analytics.
- Designed and implemented a production-grade data engineering solution for Chick-fil-A using AWS Redshift, Athena, and Apache Airflow for orchestration, automating reconciliation between multiple inventory and sales forecasting systems.
- Utilized Git version control and CI/CD practices to maintain code quality and enable seamless collaboration across cross-functional teams.
- Collaborated with multiple clients to develop and deploy custom predictive analytics solutions, including classification models, forecasting systems, and recommendation engines, translating business requirements into scalable ML products with measurable ROI.

Data Science Intern

Merkle Inc.
New York City, NY (Remote)
June 2021 - August 2021

- Engineered customer segmentation models for targeted marketing campaigns using Python and advanced statistical techniques within Databricks notebooks.
- Developed and optimized data pipelines for the 2021 Jaguar/Land Rover campaigns, performing audience overlap analysis that improved campaign targeting efficiency by 15%.
- Implemented machine learning algorithms to calculate propensity scores, working within an end-to-end MLOps framework to ensure model reproducibility and deployment readiness.
- Utilized Databricks for collaborative data engineering, feature processing, and model versioning in a production-oriented environment.

Forecasting Analyst

Genworth
Lynchburg, VA
January 2020 - March 2021

- Analyzed call center data and developed a time-series forecasting model to predict future workload and staffing requirements.
- Developed models using Python and PostgreSQL, capturing trends, seasonality, and exogenous factors affecting call volume.
- Supported quantitative research efforts in response to the COVID-19 pandemic.

Education

I hold a Bachelor of Science in Data Analytics and Data Management from Western Governors University, a Graduate Certificate in Applied Statistics and Data Management from North Carolina

State University, and a Master of Science in Geospatial Information Science and Technology from North Carolina State University.

Bachelor of Science in Data Analytics

- Western Governor's University
- Graduation: Fall, 2021

Graduate Certificate in Statistics

- North Carolina State University
- Completion: Spring, 2024

Master of Science in Geospatial Information Science and Technology

- North Carolina State University
- Graduation: Spring, 2025

Independent Research/Projects

Species Distribution Modeling

- Conducted comparative analysis between machine learning and traditional approaches (Inhomogeneous Poisson Point Process, MaxEnt) for Species Distribution Modeling, implementing all analytical pipelines in R.
- Analyzed distribution patterns of 8 bird species across 4 US states using bioclimatic variables, developing fully reproducible workflows with robust version control.

Utilizing the Clay Foundation Model and Sentinel-2 Imagery for Urban Growth Monitoring

- Developed proof-of-concept study integrating Sentinel-2 multispectral satellite imagery with the Clay Foundation Model (open-source deep Earth Observation foundation model) to monitor urban growth patterns using Python and PyTorch.
- Engineered a scalable, data-efficient framework generating urban imperviousness estimates with higher temporal resolution than standard datasets like the National Land Cover Database.
- Processed Sentinel-2 imagery into structured geospatial datasets and leveraged Clay Foundation Model for feature extraction and embedding generation.
- Designed and evaluated both simple and complex neural network architectures in PyTorch, enhancing model performance through systematic hyperparameter optimization to achieve high accuracy in urban imperviousness prediction.
- Demonstrated methodology's potential for enhancing urban monitoring frequency and responsiveness, directly supporting sustainable urban planning and resource management initiatives.

Skills

Programming Languages

- Python
- R
- SAS
- SQL/NoSQL
- Javascript/HTML/CSS

Software/Technologies

- Git
- Unix/Linux
- AWS (Athena, Redshift, EC2, Lambda)
- Dashboards (Power BI, Shiny, Dash)
- GIS Software (Esri Products, QGIS, GRASS GIS)
- MLOps Software (Dataiku, DataRobot, MLflow, Domino Data Lab)

Other Skills/Experience

- ETL and Data Integration
- Statistics & Applied Mathematics
- Machine Learning
- Deep Learning (PyTorch, TensorFlsow, Keras)
- Natural Language Processing (NLP)
- LLMs (Retrieval-Augmented Generation, LangChain)
- Forecasting
- Geospatial Analytics & Spatial Statistics

Certifications/Achievements

- AWS Cloud Practitioner, July 2024 - *Amazon Web Services (AWS)*
- IT Information Library Foundations Certification (ITIL), June 2021 - *AXELOS Global Best Practice*

Personal Attributes

With a strong work ethic and a results-driven mindset, I am committed to personal and professional growth. I thrive in dynamic environments that challenge me to push the boundaries of my knowledge and skill set. Highly organized and detail-oriented, I am passionate about continuously learning and applying new technologies and analytical approaches to solve complex problems and contribute to organizational success.